Hanyeol Lee

Robotics Researcher | Ph.D. Candidate
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Summary

I am a Ph.D. candidate in the Department of Aerospace Engineering at Seoul National University (SNU). My research vision is to enable robotic systems to perceive and understand the world at a semantic level by leveraging diverse sensing modalities, including LiDAR, monocular and stereo vision, event cameras, radar, and IMUs. Toward this goal, I integrate object-level perception with reliable state estimation to develop smarter perception systems for intelligent robots, autonomous vehicles, and aerial robotics.

Education

Ph.D. in Aerospace Engineering, Seoul National University (SNU) *2021–Present*
Department of Aerospace Engineering, Seoul, Republic of Korea

B.S. in Mechanical and Aerospace Engineering, Seoul National University (SNU) *2021*
Department of Mechanical and Aerospace Engineering, Seoul, Republic of Korea

Publications

International Journal

1. **Hanyeol Lee**, Yeongkwon Choe, Taeyoon Kim, and Chan Gook Park, "UB-SLOT: Unified Bayesian Filtering Framework for LiDAR-Inertial Simultaneous Localization and Object Tracking," *IEEE Transactions on Robotics* (*T-RO*), under review.
2. Sohee Park, **Hanyeol Lee**, Jae Hong Lee, and Chan Gook Park, "Robust Pedestrian Visual-Inertial Localization through Uncertainty-Aware Fusion with Pedestrian Dead Reckoning," *IEEE Access*, under review.
3. Taeyoon Kim, **Hanyeol Lee**, and Chan Gook Park, "Mask Refinement via DCT-Based Filtering in SAM2 for Infrared Target Tracking," *Signal, Image and Video Processing*, under review.
4. Jae Hong Lee, **Hanyeol Lee**, and Chan Gook Park, "Robust Track-to-Track Association under Sensor Asynchrony and Bias via Stochastic Trajectory Registration," *IEEE Access*, under review.
5. Taehun Yun, Youngwoo Lee, **Hanyeol Lee**, and Yeongkwon Choe, "Image-Based Learning of SD-Map-Compatible Road Centerlines for Vehicle Localization," *International Journal of Control, Automation and Systems* (*IJCAS*), under review.
6. **Hanyeol Lee**, Yeongkwon Choe, and Chan Gook Park, "Uncertainty-Aware LiDAR Object Registration with Gaussian Process in Urban Environment," *IEEE Transactions on Automation Science and Engineering* (*T-ASE*), 2025.
7. **Hanyeol Lee**, Jae Hyung Jung, Wonyoung Chung, Suyong Lee, and Chan Gook Park, "PPLIO: Plane-to-Plane LiDAR-Inertial Odometry with Multi-View Constraint in Real-Time," *IEEE Access*, 2025.
8. Byeongpil Choi, **Hanyeol Lee**, and Chan Gook Park, "Event-Frame-Inertial Odometry Using Point and Line Features Based on Coarse-to-Fine Motion Compensation," *IEEE Robotics and Automation Letters* (*RA-L*), 2025.
9. Suyong Lee, **Hanyeol Lee**, and Chan Gook Park, "Maximizing Measurement Visual-Inertial Odometry for Low-Light Environment through Online Image Pre-Processing," *IEEE Access*, 2025.
10. Sangbum Lee, **Hanyeol Lee**, and Chan Gook Park, "GRVIO: Semantic-Aware Visual-Inertial Odometry for Ground Robot Platforms," *IEEE Access*, 2025.

International Conference

1. **Hanyeol Lee**, Yeongkwon Choe, Taeyoon Kim, and Chan Gook Park, "MOTCues: 3D Multi-Object Tracking with Birth Prior and Shape Description Informed by Point Cloud Cues," *IEEE International Conference on Robotics and Automation* (*ICRA*), 2026.
2. Yerang Mok, **Hanyeol Lee**, and Chan Gook Park, "Radar-Inertial Odometry Using Line Features with Coarse-to-Fine Registration for Ground Platform," *International Conference on Indoor Positioning and Indoor Navigation* (*IPIN*), 2026.
3. Taehun Yun, Youngwoo Lee, **Hanyeol Lee**, and Yeongkwon Choe, "Learning Road Centerline Representations from Images for SD-Map Localization," *International Conference on Control, Automation, and Systems* (*ICCAS*), 2026.
4. **Hanyeol Lee** and Chan Gook Park, "Fine-Registered Object LiDAR-Inertial Odometry for a Solid-State LiDAR System," *International Conference on Artificial Life and Robotics* (*ICAROB*), 2025.
5. **Hanyeol Lee** and Chan Gook Park, "Object Pose Compensation Algorithm via Joint Shape-Texture Alignment," *International Conference on Control and Robotics* (*ICCR*), 2025.

6. Sohee Park, **Hanyeol Lee**, and Chan Gook Park, "Robust Visual-Inertial Odometry with Pedestrian Dead Reckoning under Vision-Degraded Conditions," *IEEE Sensors*, 2025.
7. Yerang Mok, **Hanyeol Lee**, and Chan Gook Park, "Radar Odometry Using 3D Feature-Assisted ICP," *International Conference on Robotics, Automation, and Artificial Intelligence (RAAI)*, 2025.
8. **Hanyeol Lee**, Jae Hyung Jung, and Chan Gook Park, "2D-3D Object Shape Alignment for Camera-Object Pose Compensation in Object-Visual SLAM," *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
9. Suyong Lee, **Hanyeol Lee**, and Chan Gook Park, "Online Image Pre-Processing Using Bayesian Optimization for Visual-Inertial Odometry," *IEEE International Conference on Automation Science and Engineering (CASE)*, 2024.
10. Byeongpil Choi, **Hanyeol Lee**, and Chan Gook Park, "Event-Based Visual-Inertial Odometry Using Point and Line Features with a Coarse-to-Fine Motion Compensation Scheme," *40th Anniversary of the IEEE International Conference on Robotics and Automation (ICRA40)*, 2024.
11. Sangbum Lee, **Hanyeol Lee**, and Chan Gook Park, "G4Q-VIO: Ground Constraint for a Quadruped Robot VIO," *40th Anniversary of the IEEE International Conference on Robotics and Automation (ICRA40)*, 2024.
12. **Hanyeol Lee**, Jae Hyung Jung, and Chan Gook Park, "LiDAR Odometry with Pre-Filtering of Plane Feature Points," *ION GNSS+*, 2023.
13. **Hanyeol Lee**, Jae Hyung Jung, Wonyoung Chung, and Chan Gook Park, "Performance Analysis of LiDAR Odometry and Mapping with Plane and Edge Features," *Asian Control Conference (ASCC)*, 2022.
14. Jae Hyung Jung, **Hanyeol Lee**, and Chan Gook Park, "Performance Analysis of Visual-Inertial Navigation System with Feature Track Parameters," *International Conference on Control, Automation, and Systems (ICCAS)*, 2022.
15. Jae Hyung Jung, **Hanyeol Lee**, and Chan Gook Park, "Visual-Inertial Navigation System Mechanized in Navigation Frame," *Asia-Pacific International Symposium on Aerospace Technology (APISAT)*, 2021.

Awards

2026 ICRA Conference Participation Award, Institute of Control, Robotics and Systems (ICROS)

2025 Best Paper Award, Institute of Control, Robotics and Systems Annual Conference (ICROS)

2024 Outstanding Paper Award, Institute of Navigation and Positioning Technology Conference (IPNT)

2023 Outstanding Paper Award, Institute of Control, Robotics and Systems Annual Conference (ICROS)

Academic Activities

2024–Present Reviewer, *IEEE Robotics and Automation Letters (RA-L)*

2024–Present Reviewer, *IEEE International Conference on Robotics and Automation (ICRA)*

Research Projects

2021–Present **Development of Indoor and Outdoor Integrated Navigation Technology for Operating in Unknown and Harsh Environments**
National Research Foundation of Korea (NRF)

2022–2026 **Seamless Estimation of Position/Attitude for Mobile Devices in Indoor and Urban Environments**
National Research Foundation of Korea (NRF)

2021 **Research on State Estimation for UAM in Urban Areas**
Hyundai Motor Company

Teaching

2026 Teaching Assistant, Advanced Filtering Theory

2026 Teaching Assistant, Aerospace Sensor System

2024 Teaching Assistant, Integrated Navigation System

2022 Teaching Assistant, Aerospace Sensor System

2021 Teaching Assistant, Aerospace Linear Algebra